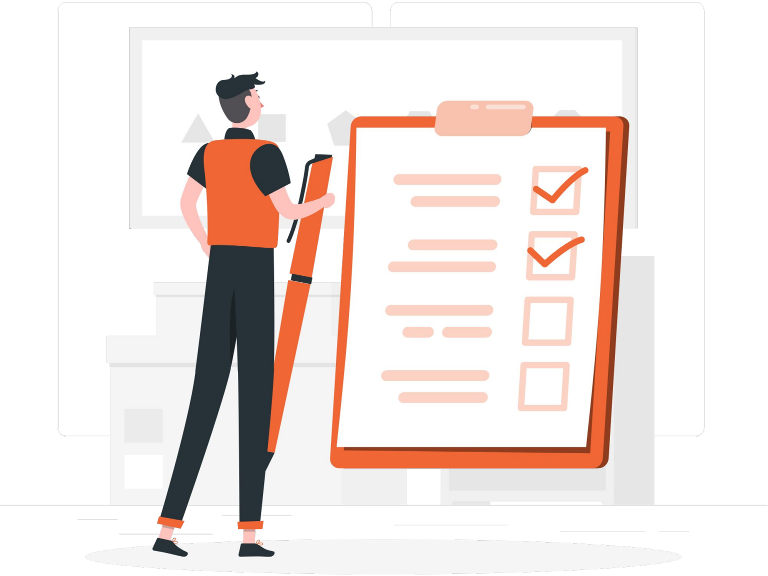
Azure Cloud Platform Overview

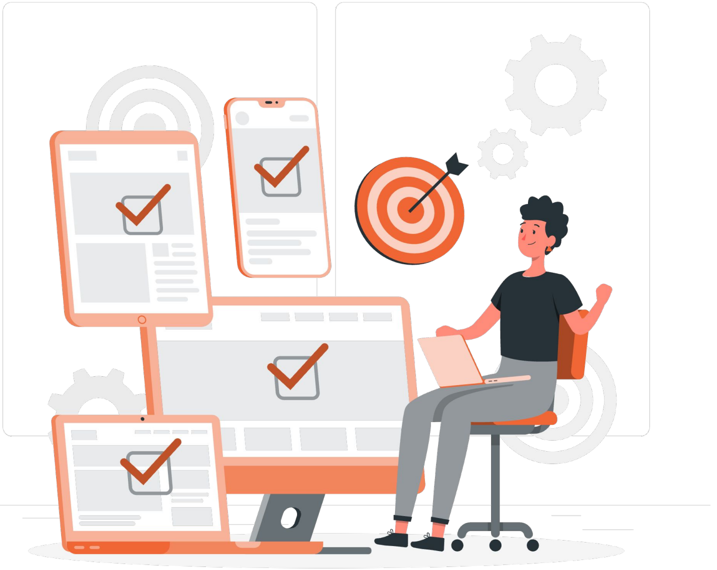
Lesson

1. Introduction to Cloud Computing
2. Introduction to Microsoft Azure
3. Core Azure Services
4. Pricing & Support



Learning Outcome

1. Master cloud computing fundamentals and deployment models.
2. Understand Azure infrastructure and resource hierarchy.
3. Identify Azure core services and use cases.
4. Manage Azure pricing, costs, and support plans.



Introduction to Cloud Computing

What is Cloud Computing

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NIST DEFINITION

"On-demand network access to a shared pool of configurable computing resources — networks, servers, storage, applications, and services — that can be rapidly provisioned and released with minimal management effort."

— National Institute of Standards and Technology

In simple terms:

Renting computing power, storage, and software over the internet — and paying only for what you use.

Think of it like electricity

On Demand
Plug in and use — no factory needed

Metered
You pay for kilowatt-hours consumed

Elastic
More appliances = more usage, automatic

Reliable
Provider handles generation & maintenance

Evolution of Computing

1960s	1980s	1990s	2000s+	2020s+
Mainframe	**Client–Server**	**Web Era**	**Cloud**	**Edge & AI**
Centralized, time-shared	Distributed PCs + servers	Internet & web apps	On-demand, pay-as-you-go	Distributed AI workloads

Key drivers of cloud adoption

- Faster time-to-market
- Cost optimization (CapEx → OpEx)
- Global scale & reach
- Built-in resilience & security

On Premise vs Cloud Computing

On Premise	
Cost Model	CapEx — buy hardware upfront
Provisioning	Weeks to months
Scalability	Manual, limited by hardware
Maintenance	In-house team responsible
Risk	Capacity over/under-provisioning
Geo Reach	Limited to physical locations

Cloud Computing	
Cost Model	OpEx — pay for usage
Provisioning	Minutes via portal or API
Scalability	Elastic, auto-scale on demand
Maintenance	Cloud provider handles it
Risk	Right-sized, pay-as-you-go
Geo Reach	Global within minutes

Six Benefits of Cloud Computing

Cost Efficiency

Pay only for resources used; no upfront capital

Global Scale

Deploy across regions in minutes

Performance

Latest hardware, low-latency networks

Speed & Agility

Provision in minutes, iterate faster

Productivity

Less rack-and-stack, more business value

Reliability & Security

Built-in redundancy and certifications

IaaS — Infrastructure as a Service

IaaS gives you the building blocks of IT — virtual machines, storage, and networks — over the internet, while the provider runs the physical hardware.

WHAT YOU MANAGE

- Operating systems & patches
- Middleware & runtime
- Applications and data
- Network configuration (within VNet)

PROVIDER MANAGES

- Physical servers, storage, networking
- Virtualization & datacenter facilities

Azure IaaS examples

Azure Virtual Machines
Windows / Linux VMs

Azure Disk Storage
Persistent block storage

Azure Virtual Network
Private cloud networking

Azure Load Balancer
Layer-4 traffic distribution

USE CASES

Lift-and-shift migrations, disaster recovery, test/dev environments, custom workloads requiring full OS control.

PaaS — Platform as a Service

PaaS provides a managed platform for developers to build, deploy, and scale applications — without worrying about servers, OS, or runtimes.

WHAT YOU MANAGE

- Application code
- Application data
- App configuration

PROVIDER MANAGES

- Runtime, middleware, OS
- Servers, storage, networking
- Patching and high availability

Azure PaaS examples

Azure App Service
Web apps, REST APIs

Azure SQL Database
Managed relational DB

Azure Functions
Serverless event handlers

Azure Kubernetes Service
Managed container orchestration

USE CASES

Modern web/mobile backends, API hosting, microservices, data services, and rapid app development with CI/CD.

SaaS — Software as a Service

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SaaS delivers complete applications to end users over the internet — typically by subscription. The provider runs everything; you just sign in and use it.

"Pizza-as-a-Service" Analogy

On-Premises	Make it from scratch at home
IaaS	Buy ingredients, cook yourself
PaaS	Order takeaway, eat at home
SaaS	Eat at the restaurant

Azure SaaS examples

Microsoft 365
Outlook, Word, Excel, Teams

Dynamics 365
ERP and CRM

Salesforce
CRM platform

Google Workspace
Productivity suite

USE CASES

Email, collaboration, CRM, ERP, accounting — anywhere ready-to-use applications meet the business need.

Cloud Deployment Models

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Public

PROS

Lowest cost, fastest scale, no infra to manage

CONS

Less direct control, shared infrastructure

Private

PROS

Full control, dedicated hardware, custom security

CONS

Higher cost, slower scaling, you operate it

Hybrid

PROS

Best of both — keep sensitive data on-prem

CONS

More complex networking and identity

Multi-Cloud

PROS

Avoid vendor lock-in, best-of-breed services

CONS

Operational and skill complexity

Shared Responsibility Model

Security and operations are split between the cloud provider and the customer. The split shifts based on the service model.

Responsibility Layer	On-Prem	IaaS	PaaS	SaaS
Information & data	Customer	Customer	Customer	Customer
Devices (mobile/PCs)	Customer	Customer	Customer	Customer
Accounts & identities	Customer	Customer	Customer	Customer
Identity & directory	Customer	Shared	Shared	Shared
Applications	Customer	Customer	Shared	Microsoft
Network controls	Customer	Customer	Shared	Microsoft
Operating system	Customer	Customer	Microsoft	Microsoft
Physical hosts	Customer	Microsoft	Microsoft	Microsoft
Physical network	Customer	Microsoft	Microsoft	Microsoft
Physical datacenter	Customer	Microsoft	Microsoft	Microsoft

Introduction to Microsoft Azure

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What is Microsoft Azure?

Microsoft's flagship public cloud

Microsoft Azure is a hyperscale, enterprise-ready public cloud platform offering compute, storage, networking, AI, analytics, IoT, and many more services consumed on demand.

Brief history

- Announced 2008 as Project Red Dog
- Launched February 2010 as Windows Azure
- Renamed Microsoft Azure in 2014
- Now offers 200+ services worldwide

200+
Services

60+
Regions

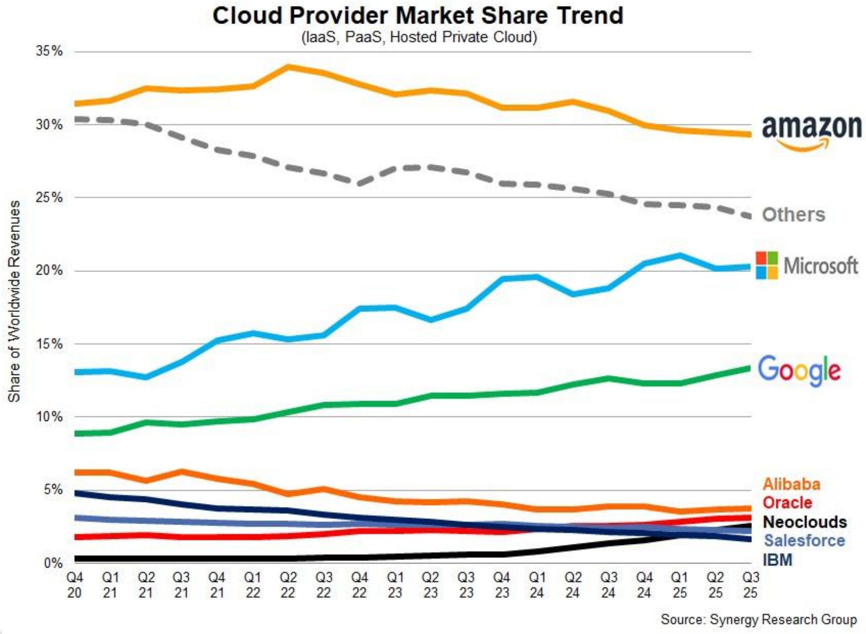
95%
Fortune 500 use Azure

300+
Data centers

Trusted by enterprises

Banking, government, healthcare, manufacturing — Azure powers mission-critical workloads with deep compliance and hybrid capabilities.

Azure in the Cloud Market



Azure's enterprise strengths

Microsoft 365 / AD integration
Native identity and productivity

Hybrid leadership
Azure Arc, Stack, and on-prem connectivity

AI & OpenAI
First-class access to GPT models

Compliance breadth
100+ certifications including PCI DSS, ISO 27001

Source: https://www.srgresearch.com/articles/cloud-market-share-trends-big-three-together-hold-63-while-oracle-and-the-neoclouds-inch-higher

Regions and Availability Zones

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Azure Region (e.g., Indonesia Central)

Availability Zone 1	Availability Zone 2	Availability Zone 3
Independent datacenter	Independent datacenter	Independent datacenter
Power Cooling Networking	Power Cooling Networking	Power Cooling Networking

Region

A geographic area containing one or more datacenters. Resources are deployed to a region.

Availability Zone (AZ)

Physically separate datacenters within a region. Independent power, cooling, and networking.

Region Pair

Two regions paired for DR (e.g., Southeast Asia + East Asia). Updates rolled sequentially.

Use Availability Zone for High Availability within a region; pair regions for Disaster Recovery across geographies.

Azure Resource Hierarchy

Management Group

Group multiple subscriptions for unified policy and access (e.g., BRI Group → Subsidiaries)

Subscription

Billing boundary; logical container for resources (e.g., Production, Dev/Test)

Resource Group

Logical group of resources sharing lifecycle (e.g., ALIA-Production-RG)

Resources

Actual services: VMs, storage accounts, databases, web apps

Each level inherits policies, RBAC, and tags from the level above.

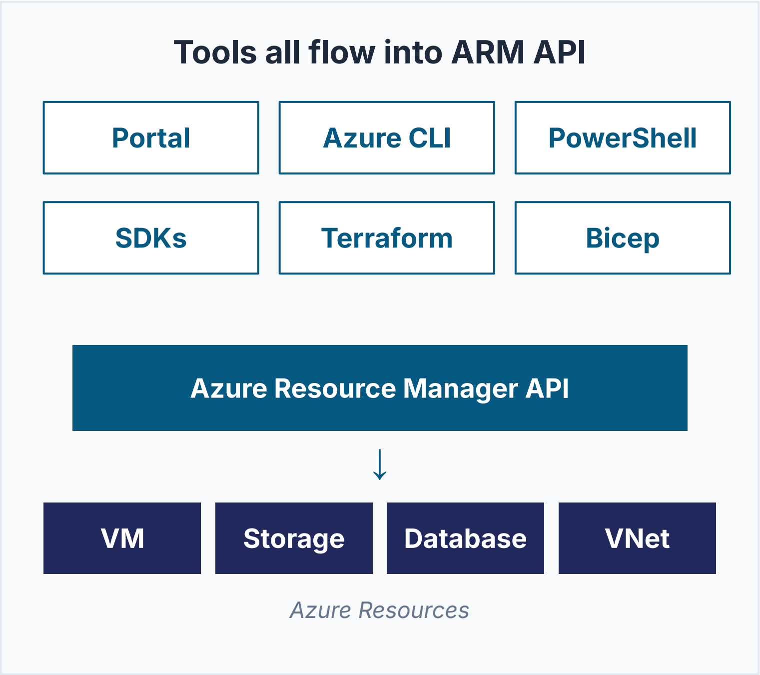
Azure Resource Manager (ARM)

The deployment & management layer

ARM is the unified API every Azure tool talks to — Portal, CLI, PowerShell, SDKs, Terraform. It enables consistent, declarative deployment of resources.

Benefits

- Declarative templates (ARM, Bicep)
- Consistent deployment across environments
- Idempotent — same template, same result
- Native dependency management
- Built-in tagging and policy support



How to Access Azure

Azure Portal

Web-based GUI for visual management. Best for beginners and one-off tasks.

Azure CLI

Cross-platform command-line. Scriptable, ideal for automation and pipelines.

Azure PowerShell

Windows-friendly cmdlets. Strong for Windows admins and complex scripts.

Azure SDKs

Programmatic access from Python, .NET, Java, JS, Go for building apps.

Azure Mobile App

iOS/Android — monitor resources, view alerts, run quick commands.

ARM / Bicep / Terraform

Infrastructure-as-code for repeatable, version-controlled deployments.

Azure Marketplace

An online store of pre-built, certified solutions from Microsoft and thousands of partners — VMs, container images, SaaS apps, AI services, and consulting offers.

20K+ Apps
Solutions to choose from

Microsoft + Partners
Vetted ecosystem

1-Click Deploy
Fast provisioning

Unified Billing
Through Azure

What's available

VM Images
OS, app stacks

Containers
Helm charts, K8s apps

SaaS Apps
Subscription software

AI/ML Models
Pre-trained, custom

Dev Tools
DevOps, security

Consulting
Partner services

Azure for Enterprises

Hybrid Capabilities

- Azure Arc — manage on-prem & multi-cloud
- Azure Stack — Azure services in your DC
- ExpressRoute — private connectivity

Compliance Coverage

- 100+ certifications: ISO 27001, SOC 1/2/3
- PCI DSS, HIPAA, FedRAMP, GDPR
- Data residency options per region

Indonesian Banking Context

- OJK regulatory framework alignment
- BSSN data sovereignty considerations
- Indonesia Central region for residency

Enterprise Agreement (EA)

- Negotiated discounts at scale
- Predictable annual commitment
- Centralized billing & cost reporting

Core Azure Services

Service Categories Overview

Compute

VMs, containers, serverless, app hosting

Storage

Blob, file, queue, disk, archive

Networking

VNets, load balancers, VPN, ExpressRoute

Databases

SQL, Cosmos DB, Postgres, MySQL

AI & ML

OpenAI, Cognitive Services, ML Studio

Analytics

Synapse, Databricks, Data Factory

Security

Defender, Sentinel, Key Vault

DevOps

Azure DevOps, GitHub, Pipelines

This deck focuses on Compute, Storage, Networking, and Databases — the foundation everyone needs.

Azure Compute Services

Compute Services Overview

Virtual Machines

IaaS — full OS control, traditional workloads

Best for: Lift-and-shift, custom apps

App Service

PaaS — managed web apps and APIs

Best for: Web/REST/mobile backends

Azure Functions

Serverless — event-driven, pay-per-execution

Best for: Glue code, automation, integration

Container Instances

Run a single container without managing infra

Best for: Quick batch jobs, simple containers

Azure Kubernetes (AKS)

Managed Kubernetes for microservices

Best for: Production microservices at scale

Container Apps

Serverless containers with built-in scaling

Best for: Modern apps without K8s ops

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Azure Virtual Machines

On-demand, scalable IaaS compute. Choose your OS, size, region, and disks — pay per second of usage.

VM Size Families

B	D	E	F	L	N
Burstable	**General Purpose**	**Memory Optimized**	**Compute Optimized**	**Storage Optimized**	**GPU**
Dev/test, low CPU avg	Balanced workloads	In-memory DBs, analytics	Web servers, gaming	Big data, NoSQL	AI/ML, rendering, HPC

Operating Systems

- Windows Server (2016, 2019, 2022, 2025)
- Linux: Ubuntu, RHEL, SUSE, Oracle, Debian
- Custom images via Azure Compute Gallery

Common Use Cases

- Lift-and-shift of existing apps
- Custom workloads needing OS control
- Dev/test environments
- SAP, Oracle, SQL Server hosting

Azure App Service

A fully-managed PaaS for hosting web apps, REST APIs, and mobile backends — without managing servers.

Supported Languages & Stacks

.NET	Java	Node.js	Python
PHP	Ruby	Go (containers)	Static HTML

Key Features

Auto-scale

Scale up/out by metric or schedule

CI/CD built-in

GitHub, Azure DevOps, Bitbucket

Deployment slots

Blue-green, swap to production

TLS & auth

Free SSL, Easy Auth (AAD, Google, FB)

Azure Functions & Serverless

Event-driven serverless compute. Write code that runs in response to triggers — pay only when it executes.

Triggers → Function → Outputs

Triggers	Function	Outputs
HTTP request		Database write
Timer (cron)		Email / SMS
Queue message	**Azure Function**	Queue message
Blob upload	*Your code runs*	Blob storage
Event Grid		API call
Service Bus		Log entry

Hosting Plans:

- Consumption (true serverless, pay-per-execution)
- Premium (warm instances, VNet)
- Dedicated (App Service plan)

Containers on Azure

Azure Container Instances (ACI)

Run a single container with no orchestration. Per-second billing.

Use when: Quick batch jobs, simple workloads, dev/test.

Azure Kubernetes Service (AKS)

Managed Kubernetes — control plane is free; you pay nodes.

Use when: Microservices at scale, complex orchestration.

Azure Container Apps

Serverless containers built on Kubernetes & Dapr — no K8s ops.

Use when: Event-driven microservices without operating K8s.

Azure Container Registry (ACR)

Private Docker registry with geo-replication and security scanning.

Use when: Storing and distributing your container images.

Azure Storage Services

Storage Services Overview

Blob Storage
Massively-scalable object storage

Backups, media, data lake

File Storage
Managed SMB / NFS shares

Lift-and-shift file shares

Queue Storage
Simple message queues

Decoupling app components

Table Storage
Schemaless NoSQL key-value

Lightweight structured data

Disk Storage
Persistent block storage for VMs

VM OS and data disks

Storage Access Tiers

Tier	Description
Hot	Frequent access — highest storage cost, lowest access cost
Cool	Infrequent access (≥30 days) — lower storage, higher access cost
Cold	Rarely accessed (≥90 days) — even lower storage cost
Archive	Almost never accessed (≥180 days) — cheapest, hours to rehydrate

Azure Blob Storage Deep Dive

Blob Types

Block Blob

Default. Text & binary data, up to ~190 TiB. Most common.

Page Blob

Random read/write 512-byte pages. Backs Azure VM disks.

Append Blob

Optimized for append-only data — logging scenarios.

Capabilities

Lifecycle Management
Auto-tier or delete objects by rules

Encryption
AES-256 at rest by default; CMK supported

Access Control
Azure AD, SAS tokens, account keys

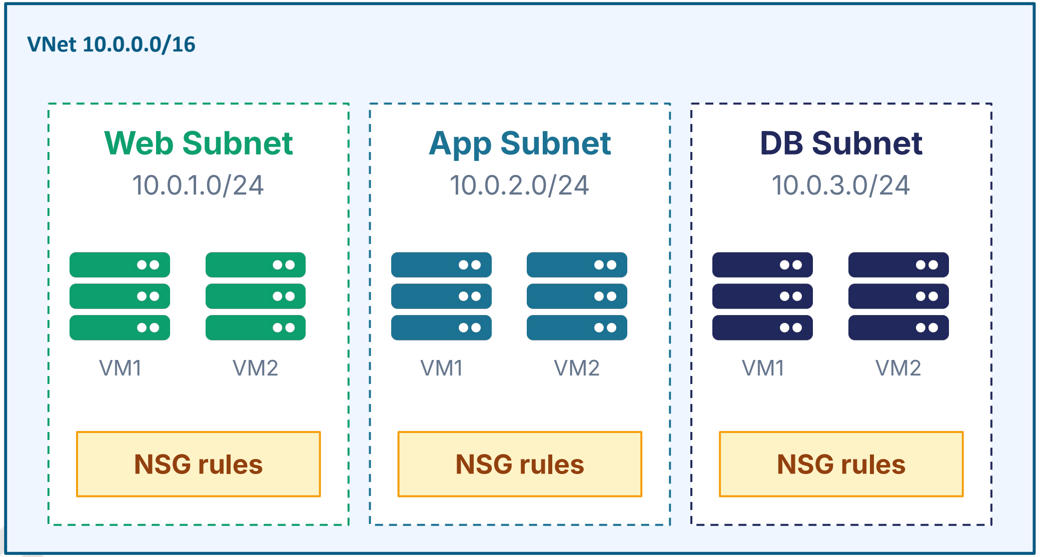
Replication
LRS, ZRS, GRS, RA-GRS for HA/DR

Hierarchical Namespace
Enable Data Lake Gen2 capabilities

Azure Storage Services

Azure Virtual Network (VNet)

VNet is your private network space in Azure. It's logically isolated, fully under your control, and the foundation for all networking.



Address Space

CIDR ranges (e.g., 10.0.0.0/16) for the VNet

Subnets

Sub-ranges to organize and isolate resources

NSGs

Allow/deny rules per subnet or NIC

VNet Peering

Connect two VNets privately, low-latency

Service Endpoints

Private routes to Azure PaaS services

Connectivity Options to Azure

Option	Type	Use Case	Bandwidth	SLA
Site-to-Site VPN	Encrypted (IPsec)	Connect on-prem network to Azure	Up to 10 Gbps	99.95%
Point-to-Site VPN	Encrypted (per device)	Individual remote workers	Lower	Best effort
ExpressRoute	Private circuit	Mission-critical, predictable latency	Up to 100 Gbps	99.95%
ExpressRoute Direct	Direct fiber	Massive scale, regulated data	Up to 100 Gbps	99.95%
Virtual WAN	Managed hub-spoke	Large multi-branch architectures	Varies	99.95%

Banking workloads typically use ExpressRoute for predictable latency, regulatory compliance, and bandwidth guarantees.

Azure Database Services

Database Services Overview

Relational	NoSQL	Cache & Analytics
Azure SQL Database Managed SQL Server	**Azure Cosmos DB** Multi-model, globally distributed	**Azure Cache for Redis** In-memory data store
SQL Managed Instance Near 100% SQL compatibility	**Cosmos DB for MongoDB** Mongo API, managed	**Synapse Analytics** Data warehouse & analytics
PostgreSQL Flexible Server Open-source Postgres	**Cosmos DB for Cassandra** Cassandra API	**Azure Databricks** Apache Spark managed
MySQL Flexible Server Open-source MySQL	**Table Storage** Simple key-value	**Data Explorer** Time-series analytics

Decision rule: relational for ACID transactions, NoSQL for global scale & flexibility, cache to accelerate hot reads.

Azure SQL Database

Fully-managed relational PaaS based on the SQL Server engine — Microsoft handles patching, backups, and HA.

Deployment Options

Single Database

One isolated DB. Simple, predictable.

Best: Modern apps, microservices

Elastic Pool

Multiple DBs share resources.

Best: SaaS multi-tenant apps

Managed Instance

Near-100% SQL Server compatibility.

Best: Lift-and-shift legacy SQL

Built-In Features

Auto Backup
PITR up to 35 days

TDE Encryption
By default

Built-in HA
99.995% SLA

Hyperscale
Up to 100 TB

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Azure SQL Database

Globally distributed, multi-model NoSQL database with single-digit millisecond latency at any scale.

Multi-Model APIs

NoSQL (native)	MongoDB	Cassandra	Gremlin (graph)	Table

Five Consistency Levels

Strong	Bounded Staleness	Session	Consistent Prefix	Eventual
Linearizable; highest consistency	Lag bounded by time or versions	Read-your-writes (default)	Reads in order of writes	No order; lowest latency

Common Use Cases: Real-time apps, IoT telemetry, personalization engines, gaming leaderboards, retail catalogs at global scale

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Pricing & Support

Azure Pricing Models

Pay-As-You-Go

Pay per second/minute of use. No commitment.

Best for: Variable, unpredictable workloads

Reserved Instances

1- or 3-year commitment for up to 72% off list.

Best for: Steady-state, predictable workloads

Azure Savings Plans

Hourly spend commitment, flexible across services.

Best for: Mixed compute workloads

Spot Pricing

Up to 90% off — but VM may be evicted.

Best for: Batch jobs, dev/test, fault-tolerant

Azure Hybrid Benefit

Bring your existing Windows / SQL licenses.

Best for: Customers with Software Assurance

Free Tier & Credits

12 months of free services + always-free items.

Best for: Learning and prototyping

Factors Affecting Cost

Resource Type & Size

VM family, DTU/vCore tier, and capacity drive base price.

Region

Pricing varies by region — US/EU often cheaper than emerging regions.

Bandwidth (Egress)

Inbound is free; outbound across regions or to internet is billed per GB.

Storage Tier

Hot vs Cool vs Archive; LRS vs GRS — different price points.

Reserved vs On-Demand

Commitments cut cost 30–72% — at the price of flexibility.

Add-On Features

Backup, monitoring, Defender, advanced threat protection — each priced.

Cost Management Tools

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1. **Azure Pricing Calculator**
   Estimate the cost of a planned architecture before deploying — free, web-based, supports all services.
2. **TCO Calculator**
   Compare current on-premises costs vs running the same workloads in Azure — quantifies migration ROI.
3. **Cost Management + Billing**
   Native Azure dashboards: spend trends, breakdowns by tag/RG/subscription, anomaly detection.
4. **Budgets & Alerts**
   Set monthly/quarterly budgets per scope; trigger email and webhook alerts before overspend.
5. **Azure Advisor**
   Personalized recommendations to right-size VMs, delete unused resources, and adopt reservations.

Azure Support Plans

Plan	Price	Severity A Response	Architecture Support	Best For
Basic	Free	—	Self-help only	Trial, learning
Developer	~$29/mo	8 business hrs	General guidance	Non-prod workloads
Standard	~$100/mo	1 hour, 24×7	General guidance	Production workloads
Pro Direct	~$1,000/mo	1 hour, 24×7	Advisory & training	Business-critical

Prices are illustrative — verify on the Azure pricing page. Severity-A = critical production-down issues.

Azure Service Level Agreements

An SLA is Microsoft's financially-backed commitment for uptime. If missed, you receive service credits.

99.99%	99.95%	99.95%	99.999%
VM in 2+ AZs	**VM in Avail. Set**	**App Service (Standard+)**	**Cosmos DB (multi-region)**
≤ 4.4 min/month	*≤ 22 min/month*	*≤ 22 min/month*	*≤ 26 sec/month*

Composite SLA

If your app uses multiple services, multiply their SLAs.
Example:
App Service (99.95%) × Azure SQL (99.99%) = 99.94%.

Service Credits

- < SLA: 10% credit on monthly bill
- < 99.0%: 25% credit
- Below 95%: 100% credit (typical)

Guided Hands-on

Dealer Sales Reporting Portal

The Scenario

Berlian Sistem Informasi needs an internal web portal where Mitsubishi dealer staff across Indonesia can submit weekly sales reports and view consolidated dashboards.

The portal must be accessible from any dealer office during business hours, store sales documents, and authenticate users via existing corporate identity.

KEY NUMBERS

~50	Mitsubishi dealers
~200	concurrent users (peak)
8 hr	business hours/day
<5 GB	data per month

YOUR GOAL *Design the Azure architecture for this portal and estimate the monthly running cost.*

Architecture Design

Dealer Staff
~200 concurrent

HTTPS

WEB TIER
Azure App Service
Plan: B1 (Basic)
Web app hosting

→

DATA TIER
Azure SQL Database
Tier: Standard S0
Sales records

↓

STORAGE
Blob Storage
Reports & PDFs

IDENTITY
Microsoft Entra ID
Single sign-on

Design Decisions

Why App Service?
PaaS — no server management, built-in scaling, easy CI/CD

Why SQL Database?
Sales data is relational; managed backups, no DBA overhead

Why Blob Storage?
Cheap, scalable storage for PDF reports and images

Why Entra ID?
Reuse existing Mitsubishi corporate identities

Region: Indonesia Central
Jakarta — in-country data residency, 3 AZs

Pricing Calculator

OPEN https://azure.microsoft.com/en-us/pricing/calculator/

1 Add App Service

Region: Indonesia Central
Tier: Basic B1
Instances: 1
Hours: 730/mo

2 Add SQL Database

Region: Indonesia Central
Deployment: Single DB
Tier: Standard S0 (10 DTU)
Storage: 250 GB included

3 Add Storage Account

Region: Indonesia Central
Type: Blob
Tier: Hot
Capacity: 50 GB
Write ops: low

4 Add Bandwidth

Outbound (egress)
Estimate: 10 GB/month
First 100 GB has
free allowance

5 Review & Adjust

Verify currency (USD)
Review each line item
Check region consistency
Add support if needed

6 Save & Share

Click Save Estimate
Name: BSI-Dealer-Portal
Export as Excel/PDF
Share URL with team

Estimated Cost Breakdown

Indicative monthly cost — Indonesia Central (in-country residency), Linux, pay-as-you-go. Verified Nov 2026 against published rates. Always re-check in the Azure Pricing Calculator before publishing.

Service	Configuration	Monthly Cost (USD)
App Service Plan	Basic B1 Linux, 1 instance, 730 hrs · Indonesia Central	$52.01
Azure SQL Database	Standard S0, 10 DTU, 250 GB included · Indonesia Central	$14.72
Blob Storage	Hot tier, LRS, 50 GB + standard ops · Indonesia Central	$1.30
Microsoft Entra ID	Free tier (basic SSO, ≤50K MAU) · Global service	$0.00
Bandwidth (egress)	10 GB — first 100 GB/mo is free · Global	$0.00
Azure Monitor	Basic metrics + 5 GB log ingest · Indonesia Central	$2.30
ESTIMATED TOTAL	*Indonesia Central (regional services) + global services*	**≈ $70.33**

Assignment

Dealer Sales Reporting Portal

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The Scenario

Mitsubishi Motors Indonesia wants to launch a customer-facing platform where car owners book service appointments online, upload photos of issues, receive real-time updates, and rate dealers after service.

Key challenges:

- Nationwide reach with consistent low latency
- Photo uploads from mobile (variable network)
- Real-time push notifications for status updates
- Integration with existing dealer management systems
- Compliance with Indonesian data residency rules

KEY NUMBERS

500k	registered users
50k	monthly active users
5k	concurrent at peak
99.95%	availability target
<2s	page load (P95)
24×7	operating hours

Requirements

FUNCTIONAL REQUIREMENTS	
F1	User registration & profile management
F2	Service appointment booking & rescheduling
F3	Photo upload (issues, vehicle damage)
F4	Real-time notifications (SMS / push / email)
F5	Dealer schedule & capacity management
F6	Post-service rating & feedback
F7	Management analytics dashboard
F8	Integration with dealer management system

NON-FUNCTIONAL REQUIREMENTS	
N1	Availability ≥ 99.95%
N2	P95 page load < 2 seconds
N3	Auto-scaling to handle 5K concurrent
N4	Encryption at rest and in transit
N5	Daily automated backups, 30-day retention
N6	Disaster recovery (RTO < 4h, RPO < 1h)
N7	Indonesian data residency where possible
N8	Centralized monitoring & alerting

Tasks & Deliverables

1. **Architecture Diagram**
   Draw a clear diagram showing all Azure services, data flows, and external integrations. Use any tool — diagrams.net, Lucidchart, or PowerPoint shapes.

2. **Service Selection & Justification**
   List every Azure service you used and explain in 1–2 sentences why you chose it over alternatives (e.g., why AKS instead of App Service?).

3. **High Availability & DR Strategy**
   Describe how your design meets the 99.95% SLA and the RTO/RPO targets. Identify single points of failure and how you mitigate them.

4. **Cost Estimation**
   Use Azure Pricing Calculator to estimate monthly cost. Apply at least one cost-optimization technique (Reserved Instance, savings plan, autoscale schedule, etc.).

5. **Trade-offs & Risks**
   Document at least three trade-offs you made (e.g., cost vs. performance, complexity vs. control) and the risks remaining in your design.

Architecture Hints

Edge & Security

- Azure Front Door
- Web Application Firewall
- Azure CDN

Compute Tier

- App Service / AKS
- Container Apps
- Functions for async work

Data Tier

- Azure SQL or Cosmos DB
- Azure Cache for Redis
- Storage for photos

Identity & Access

- Microsoft Entra External ID
- RBAC roles
- Key Vault for secrets

Messaging & Notifications

- Service Bus / Event Grid
- Notification Hubs
- Email/SMS via Comm. Svcs

Observability

- Application Insights
- Log Analytics
- Azure Monitor + alerts

Evaluation and Submission

Evaluation Criteria

Architecture Soundness	Services chosen appropriately, design covers all functional requirements	30%
Non-Functional Coverage	HA, DR, scaling, security explicitly addressed	25%
Cost Estimation	Realistic numbers, optimization technique applied	20%
Justification & Trade-offs	Clear reasoning, trade-offs documented	15%
Clarity of Diagram	Diagram is readable and self-explanatory	10%

Format

Single PDF (max 10 pages) OR shareable Azure Calculator link + 1-page architecture PDF

Deadline

Submit before next training session

Submit to

Hacktiv8 LMS or via your instructor's email

Optional bonus

Include an ARM/Bicep snippet for any one component

Thank You

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